

Foundations of Arithmetic Power Geometry VII: Toward the APG–Arakelov Projection Theorem

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Abstract

We establish the arithmetic-geometric extension of Arithmetic Power Geometry (APG) over regular arithmetic models by targeting the central projection bridge isolated in Volume VI. Moving beyond localized thermodynamic descriptions, this paper formalizes the relationship between the continuous, path-integrated deformation energy $\mathcal{E}_{\text{APG}}$ on the compactified modular curve $X_0(N)^*$ and the discrete arithmetic metrics of a projected semi-stable Frey scheme $E_{a,b,p}$. We prove the Intrinsic Discrepancy–Source Compatibility Theorem, anchoring the continuous potential directly to the positive curvature landscape of the Arakelov metric discrepancy bundle. By applying Stein factorization to graph closures and formalizing the metric self-pairing within the arithmetic Picard group, we map analytic field variations to global intersection invariants. We show that the global projection inequality is reduced to three explicit boundary conditions governing weighted cusp widths, vertical component intersection matrices, and Green-energy metric distortion fields. This completes the structural architecture of the conditional projection program and localizes the remaining analytic and arithmetic limits of the APG method.

Keywords: Arithmetic Power Geometry; Arakelov intersection theory; modular parametrization; Stein factorization; Deligne pairing; Green-energy distortion.

1 Introduction and Architectural Blueprint

Classical arithmetic geometry characterizes the distribution of rational or integral points on algebraic varieties by fixing polynomial exponent parameters while varying the underlying field of coordinates (Bombieri & Gubler, 2006; Hindry & Silverman, 2000; Lang, 1983). Arithmetic Power Geometry (APG) introduces an inverted operational paradigm: it stabilizes the positive integer baseline coordinates and treats algebraic power exponents as continuous deformation parameters over a specialized parameter space (Akhtar, 2026a, 2026c).

Early continuous formulations of APG faced metric-collapse anomalies and definitional ambiguities regarding number-theoretic conductor functions over non-integral spaces (Akhtar, 2026d, 2026e). Volume V regularized these deformation pathways by discretizing the parameter landscape over an ordered prime chain

$$S_P = \{3, 5, 7, \dots, p\}$$

on the universal moduli stack of elliptic curves $\mathcal{M}_{\text{ell}}$ (Akhtar, 2026e).

Volume VI established a conditional framework linking the path-integrated closure-defect functional to the stable Faltings height of a semi-stable Frey curve $E_{a,b,p}$ associated with a hypothetical primitive Fermat solution (Akhtar, 2026f; Frey, 1986). However, that projection link remained contingent on an unproved projection hypothesis. The core objective of APG VII is to map this missing analytic-to-arithmetic bridge using Arakelov intersection theory (Arakelov, 1974; Faltings, 1984). Rather than introducing ungrounded global assertions, this manuscript establishes an internal, regularized reduction model that identifies the precise arithmetic and analytic boundary profiles of continuous exponent deformations.

The structural flow of the paper is as follows:

$$\begin{array}{c}
\text{Discrepancy complex } \bar{L} \simeq \bar{\omega}_{\mathcal{X}} \otimes (\pi^* \bar{\omega}_{\mathcal{E}})^{-1} \\
\Downarrow \\
\text{Archimedean potential } \phi \\
\Downarrow \\
\alpha_{\text{APG}} = \mathcal{D}_p^*(a, b)H(\widetilde{W}), \quad \rho_{\phi} = \frac{(\Delta\phi)_+}{m_+} \\
\Downarrow \\
\Delta u_{\text{APG}} = \alpha_{\text{APG}} \left(\rho_{\phi} - \frac{1}{V_X} \right) \\
\Downarrow \\
\mathcal{E}_{\text{APG}} = C_{\phi} [\mathcal{D}_p^*(a, b)H(\widetilde{W})]^2 \\
\Downarrow \\
\mathcal{E}_{\text{APG}} \leq 4\pi \deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2 + O(\deg(\pi) \log N) \quad \text{conditionally.}
\end{array}$$

Proposition 1 (Conditional Metric Variation Bridge Formula). *Let $\pi : \mathcal{X} \rightarrow \mathcal{E}$ be a modular parametrization of regular arithmetic models, and let u_{APG} be the smooth mean-zero potential generating the continuous deformation field. Under the Deligne–Arakelov conformal variation normalization used in Volume VI, the relationship between $\mathcal{E}_{\text{APG}}$ and the change in the arithmetic self-intersection of the relative canonical sheaf satisfies*

$$\mathcal{E}_{\text{APG}} \leq 4\pi \left(\widehat{c}_1(\bar{\omega}_{\mathcal{X}})^2 - \deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2 \right) + O(\deg(\pi) \log N),$$

provided the boundary metric discrepancy is controlled at conductor scale.

Proof. Let $\bar{\omega}_{\text{can}}$ denote the canonical Arakelov metric on the relative dualizing sheaf $\omega_{\mathcal{X}}$, and let

$$\bar{\omega}_{\text{APG}} = \bar{\omega}_{\text{can}} e^{-u_{\text{APG}}}$$

represent the conformally twisted metric variation on the smooth complex generic fiber

$$X = X_0(N)^*(\mathbb{C}).$$

By the metric variation formula for arithmetic surfaces, a conformal twist gives

$$\widehat{c}_1(\bar{\omega}_{\text{APG}})^2 = \widehat{c}_1(\bar{\omega}_{\text{can}})^2 + \frac{1}{4\pi} \int_X |\nabla u_{\text{APG}}|^2 d\mu - 2 \int_X u_{\text{APG}} \text{ch}_1(\bar{\omega}_{\text{can}}).$$

The APG Poisson normalization imposes

$$\int_X u_{\text{APG}} d\mu = 0.$$

With the canonical Arakelov normalization, the Chern form is a fixed scalar multiple of the normalized volume form, so the linear term vanishes:

$$\int_X u_{\text{APG}} \text{ch}_1(\bar{\omega}_{\text{can}}) = 0.$$

Thus

$$\mathcal{E}_{\text{APG}} = \int_X |\nabla u_{\text{APG}}|^2 d\mu = 4\pi (\widehat{c}_1(\bar{\omega}_{\text{APG}})^2 - \widehat{c}_1(\bar{\omega}_{\text{can}})^2).$$

The projection formula for finite morphisms gives

$$\widehat{c}_1(\pi^* \bar{\omega}_{\mathcal{E}})^2 = \deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2.$$

The difference between $\bar{\omega}_{\text{APG}}$, $\bar{\omega}_{\mathcal{X}}$, and $\pi^* \bar{\omega}_{\mathcal{E}}$ is governed by horizontal ramification, vertical correction, and archimedean metric distortion. If these boundary terms are bounded by $O(\deg(\pi) \log N)$, the stated inequality follows. This is why the formula is stated as conditional at the boundary-estimate level. $\square$

2 Intrinsic Source Construction and Compatibility

A primary structural vulnerability in early APG iterations was the external specification of the deformation source term ρ , which lacked an intrinsic connection to the geometry of the target arithmetic surface. Theorem 1 resolves this configuration ambiguity by deriving the source profile directly from the curvature properties of the underlying Arakelov discrepancy object.

Theorem 1 (APG Discrepancy–Source Compatibility). *Let*

$$\bar{L} = \bar{\omega}_{\mathcal{X}} \otimes (\pi^* \bar{\omega}_{\mathcal{E}})^{-1}$$

define the Arakelov discrepancy line bundle inside the arithmetic Picard group $\text{Pic}(\mathcal{X})$ over the regular arithmetic model $\mathcal{X} = \mathcal{X}_0(N)^$. Let its archimedean metric discrepancy be represented by a smooth, mean-zero real potential ϕ on the compact Riemann surface $X = X_0(N)^*(\mathbb{C})$. Suppose*

$$m_+ = \int_X (\Delta\phi)_+ d\mu > 0,$$

and define the intrinsic APG source density by

$$\rho_\phi = \frac{(\Delta\phi)_+}{m_+}.$$

Let u_{APG} be the unique mean-zero solution of

$$\Delta u_{\text{APG}} = \alpha_{\text{APG}} \left(\rho_\phi - \frac{1}{V_X} \right), \quad \int_X u_{\text{APG}} d\mu = 0,$$

where

$$\alpha_{\text{APG}} = \mathcal{D}_p^*(a, b) H(\widetilde{W}).$$

Then

$$\mathcal{E}_{\text{APG}} = C_\phi [\mathcal{D}_p^*(a, b) H(\widetilde{W})]^2,$$

where

$$C_\phi = \int_X |\nabla \psi_\phi|^2 d\mu$$

and ψ_ϕ is the unique mean-zero solution of

$$\Delta \psi_\phi = \rho_\phi - \frac{1}{V_X}.$$

Proof. Since X is compact and ρ_ϕ is normalized, one has

$$\int_X \left(\rho_\phi - \frac{1}{V_X} \right) d\mu = \int_X \frac{(\Delta \phi)_+}{m_+} d\mu - \frac{1}{V_X} \int_X d\mu = 1 - 1 = 0.$$

The Fredholm alternative for the Laplace–Beltrami operator on compact manifolds gives a unique smooth mean-zero solution ψ_ϕ of

$$\Delta \psi_\phi = \rho_\phi - V_X^{-1}.$$

By linearity of Δ , the scaled APG solution is

$$u_{\text{APG}} = \alpha_{\text{APG}} \psi_\phi.$$

Therefore,

$$\mathcal{E}_{\text{APG}} = \int_X |\nabla u_{\text{APG}}|^2 d\mu = \alpha_{\text{APG}}^2 \int_X |\nabla \psi_\phi|^2 d\mu = C_\phi \alpha_{\text{APG}}^2.$$

Substituting

$$\alpha_{\text{APG}} = \mathcal{D}_p^*(a, b) H(\widetilde{W})$$

yields

$$\mathcal{E}_{\text{APG}} = C_\phi [\mathcal{D}_p^*(a, b) H(\widetilde{W})]^2.$$

Thus the APG source is anchored directly to the positive part of the Arakelov discrepancy curvature by construction. $\square$

3 Proved Structural Core and Remaining Boundary Estimates

To transition the continuous framework to regular arithmetic surfaces, rational maps on the generic fiber must be extended across the base scheme $\text{Spec } \mathbb{Z}$. Lemma 1 uses graph closures,

normalization, and Stein factorization to construct a morphism over regular models.

Lemma 1 (Extension by Graph Closure and Stein Factorization). *Let*

$$\pi_\eta : X_0(N)_\eta^* \rightarrow E_\eta$$

be the modular parametrization on the generic fiber. After replacing $X_0(N)^$ and E by regular arithmetic models $\mathcal{X}$ and $\mathcal{E}$, the rational map extends after graph closure, normalization, Stein factorization, and regularization to a finite morphism*

$$\tilde{\pi} : \tilde{\mathcal{X}} \rightarrow \mathcal{E}$$

from a regular arithmetic surface $\tilde{\mathcal{X}}$ birational to $\mathcal{X}$. Any vertical divisor correction is supported over primes $q \mid N$.

Proof. Let

$$\mathcal{G}_\eta \subset X_\eta \times E_\eta$$

be the schematic graph of π_η . Let $\bar{\mathcal{G}}$ be the schematic closure of $\mathcal{G}_\eta$ inside

$$\mathcal{X} \times_{\text{Spec } \mathbb{Z}} \mathcal{E}.$$

Because $\mathcal{X}$ and $\mathcal{E}$ are proper over $\text{Spec } \mathbb{Z}$, the natural projections from $\bar{\mathcal{G}}$ are proper. Let $\mathcal{G}^\nu$ be the normalization of $\bar{\mathcal{G}}$. The first projection

$$p_1 : \mathcal{G}^\nu \rightarrow \mathcal{X}$$

is proper and birational, and it is an isomorphism on the generic fiber. Applying Stein factorization gives

$$\mathcal{G}^\nu \rightarrow \mathcal{Y} \rightarrow \mathcal{X},$$

where $\mathcal{G}^\nu \rightarrow \mathcal{Y}$ has geometrically connected fibers and $\mathcal{Y} \rightarrow \mathcal{X}$ is finite. Replacing $\mathcal{X}$ by a regularization of $\mathcal{Y}$ resolves the remaining indeterminacy. The second projection then yields

$$\tilde{\pi} : \tilde{\mathcal{X}} \rightarrow \mathcal{E}.$$

Away from primes $q \mid N$, the modular level structure and the semi-stable Frey scheme have good reduction relative to the conductor. Hence no additional vertical correction is generated away from the support of N . Thus all vertical corrections are supported over $q \mid N$. $\square$

Lemma 2 (Arithmetic Riemann–Hurwitz with Explicit Metric Self-Pairing). *In the arithmetic Picard group of the regularized model, the relative dualizing sheaves satisfy*

$$\bar{\omega}_{\mathcal{X}} \cong \pi^* \bar{\omega}_{\mathcal{E}} \otimes \mathcal{O}(\bar{R}_\pi + \bar{V}_\pi),$$

where $\bar{R}_\pi$ is the horizontal ramification divisor and $\bar{V}_\pi$ is the vertical correction divisor. Moreover,

the archimedean metric discrepancy field

$$M_\pi = \log \left(\frac{\|\cdot\|_{\bar{\omega}_{\mathcal{X}}}}{\|\cdot\|_{\pi^* \bar{\omega}_{\mathcal{E}}}} \right)$$

contributes to the arithmetic self-intersection through the Dirichlet self-pairing

$$\langle \bar{M}_\pi, \bar{M}_\pi \rangle_{\text{Ar}} = \int_X |\nabla M_\pi|^2 d\mu.$$

Proof. On the smooth generic fiber, classical Riemann–Hurwitz gives

$$K_X = \pi^* K_E + R_\pi.$$

Extending this identity to regular arithmetic surfaces introduces vertical correction components supported on singular reduction fibers. Hence, in the arithmetic Picard group,

$$\bar{\omega}_{\mathcal{X}} \cong \pi^* \bar{\omega}_{\mathcal{E}} \otimes \mathcal{O}(\bar{R}_\pi + \bar{V}_\pi).$$

After equipping the relative dualizing sheaves with Arakelov metrics, the quotient metric defines a smooth real-valued discrepancy function M_π on the complex fiber. For a metrized trivial factor represented by M_π , the archimedean component of the arithmetic intersection pairing is the Dirichlet form

$$\int_X |\nabla M_\pi|^2 d\mu.$$

This gives the claimed self-pairing formula. □

Boundary Estimate A: The Cuspidal Width Boundary Estimate

To resolve the first geometric boundary gap, the compactifying divisor contribution $\bar{V}_{\text{cusp}}$ adjoined during the proper regular closure of the modular surface must be bounded by the covering degree. Let c index the cusps of $X_0(N)^*(\mathbb{C})$, and let $w(c)$ denote the cusp width under $\Gamma_0(N)$. If a_c denotes the boundary ramification weight at c , the required estimate is

$$\widehat{\deg}(\bar{V}_{\text{cusp}}) = \sum_c a_c \log w(c) \leq C \deg(\pi) \log N.$$

It is enough to prove

$$\sum_c a_c \leq C \deg(\pi), \quad w(c) \leq N \quad \text{for every cusp } c,$$

together with the corresponding weighted logarithmic width estimate. Until this weighted summation is established, the cuspidal contribution remains a boundary estimate rather than a proved lemma.

Boundary Estimate B: The Vertical Conductor Matrix Estimate

To resolve the second arithmetic gap, the vertical correction component $\bar{V}_{\text{bad}}$ supported over primes of singular reduction must be shown to scale with conductor valuations. For each prime $q \mid N$, let $\mathcal{X}_q$ denote the special fiber. The required local estimate is

$$\widehat{\deg}(V_q) \leq C \deg(\pi) v_q(N) \log q.$$

Then

$$\sum_{q \mid N} \widehat{\deg}(V_q) \leq C \deg(\pi) \sum_{q \mid N} v_q(N) \log q = C \deg(\pi) \log N.$$

Until this localized intersection-matrix inequality is verified, the vertical bad-fiber term remains a boundary estimate.

Boundary Estimate C: The Metric Distortion Green-Energy Estimate

The remaining analytic boundary gap is the metric distortion estimate. Let

$$M_{\text{dist}} = \log \left(\frac{\|\cdot\|_{\bar{\omega}_{\mathcal{X}}}}{\|\cdot\|_{\pi^* \bar{\omega}_{\mathcal{E}}}} \right)$$

be the global metric distortion field. The required estimate is

$$\int_X |\nabla M_{\text{dist}}|^2 d\mu \leq C \deg(\pi) \log N.$$

This requires degeneration estimates for canonical Green functions near logarithmic boundary components and cuspidal sectors. Until this analytic estimate is verified, the metric distortion term remains a boundary estimate.

Lemma 3 (Conditional Self-Intersection Structural Comparison). *Assume Boundary Estimates A, B, and C. Then*

$$\widehat{c}_1(\bar{\omega}_{\mathcal{X}})^2 \leq 2 \deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2 + O(\deg(\pi) \log N).$$

Proof. Expanding the arithmetic decomposition of Lemma 2 inside the arithmetic intersection ring gives

$$\widehat{c}_1(\bar{\omega}_{\mathcal{X}})^2 = \widehat{c}_1(\pi^* \bar{\omega}_{\mathcal{E}})^2 + 2 \widehat{c}_1(\pi^* \bar{\omega}_{\mathcal{E}}) \cdot (\bar{R}_{\pi} + \bar{V}_{\pi}) + (\bar{R}_{\pi} + \bar{V}_{\pi})^2 + \text{extmetricterms}.$$

The projection formula gives

$$\widehat{c}_1(\pi^* \bar{\omega}_{\mathcal{E}})^2 = \deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2.$$

Boundary Estimates A and B control the horizontal cuspidal and vertical contributions, while Boundary Estimate C controls the archimedean metric term. Under these assumptions, the remaining cross-terms and discrepancy self-intersections are absorbed into

$$\deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2 + O(\deg(\pi) \log N).$$

Therefore

$$\widehat{c}_1(\bar{\omega}_{\mathcal{X}})^2 \leq 2 \deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2 + O(\deg(\pi) \log N).$$

□

4 Conditional APG–Arakelov Projection Proof

Theorem 2 (The Conditional APG–Arakelov Projection Theorem). *Assume Boundary Estimates A, B, and C. Then the path-integrated Dirichlet deformation energy is bounded above by the modular parametrization complexity of the semi-stable Frey curve:*

$$\mathcal{E}_{\text{APG}} \leq 4\pi \deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2 + O(\deg(\pi) \log N).$$

Proof. By the conditional metric variation bridge,

$$\mathcal{E}_{\text{APG}} \leq 4\pi (\widehat{c}_1(\bar{\omega}_{\mathcal{X}})^2 - \deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2) + O(\deg(\pi) \log N).$$

By Lemma 3,

$$\widehat{c}_1(\bar{\omega}_{\mathcal{X}})^2 \leq 2 \deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2 + O(\deg(\pi) \log N).$$

Substituting gives

$$\mathcal{E}_{\text{APG}} \leq 4\pi (2 \deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2 - \deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2) + O(\deg(\pi) \log N),$$

and hence

$$\mathcal{E}_{\text{APG}} \leq 4\pi \deg(\pi) \widehat{c}_1(\bar{\omega}_{\mathcal{E}})^2 + O(\deg(\pi) \log N).$$

□

5 Honest Status and Strategy Outlook

The present manuscript proves intrinsic APG source compatibility and reduces the APG–Arakelov projection theorem to three explicit arithmetic-geometric boundary estimates. It does not prove the full projection theorem unconditionally. The conceptual advancement consists of anchoring the continuous field equations to the native Arakelov discrepancy curvature, transforming the APG method from an interpretive dictionary into a rigorous conditional reduction framework.

6 Explicit Remaining Technical Program

To bridge the remaining analytic and arithmetic gaps between continuous APG potential fields and classical height limits, the APG program isolates the following three technical problems.

Problem 1 (Resolution of Cusp Ramification Weights). *Establish a complete combinatorial mapping of the boundary stabilizers under $\Gamma_0(N)$. In particular, define the local projection*

weights a_c precisely and prove

$$\sum_c a_c \leq C \deg(\pi), \quad \sum_c a_c \log w(c) \leq C \deg(\pi) \log N.$$

Problem 2 (Explicit Special Fiber Reduction Matrices). *For every prime $q \mid N$, localize the intersection pairing over $\mathcal{X}_q$ and prove*

$$\widehat{\deg}(V_q) \leq C \deg(\pi) v_q(N) \log q.$$

This requires explicit bounds for component multiplicities and intersection matrices in the regularized semi-stable model.

Problem 3 (Green-Function Degeneration and Metric Distortion). *Develop a perturbation framework for canonical Green functions near cuspidal and logarithmic boundary regions, proving*

$$\int_X |\nabla M_{\text{dist}}|^2 d\mu \leq C \deg(\pi) \log N.$$

Completion of this estimate would remove the main analytic obstruction to the unconditional APG–Arakelov projection theorem.

7 Conclusion

This paper develops the seventh stage of the Arithmetic Power Geometry (APG) program by establishing a structural bridge between continuous APG deformation energies and arithmetic intersection theory on regular arithmetic models. The central achievement is the Intrinsic Discrepancy–Source Compatibility Theorem, which demonstrates that the APG source term can be derived canonically from the positive curvature component of the Arakelov discrepancy bundle rather than being imposed externally. As a consequence, the APG deformation energy acquires a direct geometric interpretation in terms of curvature-driven arithmetic variation.

To support this connection, the paper extends modular parametrizations to regular arithmetic models through graph closure, normalization, Stein factorization, and regularization procedures. In addition, an arithmetic Riemann–Hurwitz framework with explicit metric self-pairing is established, providing the structural decomposition necessary to compare APG energies with Arakelov intersection quantities.

The principal result of APG VII is a rigorous reduction of the APG–Arakelov Projection Problem to three explicit boundary estimates: the Cuspidal Width Boundary Estimate, the Vertical Conductor Matrix Estimate, and the Metric Distortion Green-Energy Estimate. Under these estimates, the Conditional APG–Arakelov Projection Theorem follows and yields an upper bound relating APG deformation energy to arithmetic self-intersection invariants of the projected semi-stable Frey curve.

Accordingly, APG VII does not claim an unconditional proof of the projection theorem. Rather, it identifies and isolates the precise arithmetic and analytic obstructions that must be resolved for completion of the projection program. This transforms the APG–Arakelov correspondence

from a heuristic framework into a mathematically structured conditional reduction program.

Future work will focus on establishing the remaining boundary estimates, particularly the Green-function degeneration and metric distortion problem, which presently constitutes the strongest analytical obstacle. Resolution of these estimates would determine whether the APG projection framework can be elevated from a conditional reduction theorem to a complete arithmetic-geometric projection theorem.

References

- Abbes, A., & Ullmo, E. (1997). *À propos de la conjecture de Manin pour les courbes elliptiques*. *Compositio Mathematica*, 105(3), 269–281.
- Ahlfors, L. V. (1979). *Complex analysis* (3rd ed.). McGraw–Hill.
- Akhtar, M. A. K. (2026a). *Foundations of Arithmetic Power Geometry: Local parameter deformations and entropy at the Euclidean target*. UMU Preprint Series, I, 1–15.
- Akhtar, M. A. K. (2026b). *Foundations of Arithmetic Power Geometry II: Information-geometric formulations, coordinate stability, and logarithmic projective height mappings*. UMU Preprint Series, II, 1–12.
- Akhtar, M. A. K. (2026c). *Arithmetic Power Geometry III: The integrated closure defect functional and the conditional Frey-curve height obstruction program*. UMU Preprint Series, III, 1–9.
- Akhtar, M. A. K. (2026d). *Foundations of Arithmetic Power Geometry IV: The coordinate-dependent defect complex and the modularity barrier obstruction program*. UMU Preprint Series, IV, 1–10.
- Akhtar, M. A. K. (2026e). *Foundations of Arithmetic Power Geometry V: Discrete moduli-stack formulations, scale-dependent metrics, and boundary obstruction limits*. UMU Preprint Series, V, 1–8.
- Akhtar, M. A. K. (2026f). *Foundations of Arithmetic Power Geometry VI: A conditional APG–Arakelov height coupling program*. UMU Preprint Series, VI, 1–20.
- Arakelov, S. Y. (1974). An intersection theory for divisors on an arithmetic surface. *Mathematics of the USSR-Izvestiya*, 8(6), 1167–1180.
- Atiyah, M. F., & Macdonald, I. G. (1969). *Introduction to commutative algebra*. Addison–Wesley.
- Bombieri, E., & Gubler, W. (2006). *Heights in Diophantine geometry*. Cambridge University Press.
- Bosch, S., Lütkebohmert, W., & Raynaud, M. (1990). *Néron models*. Springer.
- Bugeaud, Y. (2004). *Approximation by algebraic numbers*. Cambridge University Press.
- Cover, T. M., & Thomas, J. A. (2006). *Elements of information theory* (2nd ed.). Wiley.

- Darmon, H., Diamond, F., & Taylor, R. (1997). Fermat's Last Theorem. *Current Developments in Mathematics, 1995*(1), 1–154.
- Deligne, P. (1987). Le déterminant de la cohomologie. In *Current trends in arithmetical algebraic geometry* (pp. 93–177). American Mathematical Society.
- Diamond, F. (1996). On deformation rings and Hecke rings. *Annals of Mathematics, 144*(1), 137–166.
- Diamond, F., & Shurman, J. (2005). *A first course in modular forms*. Springer.
- Faltings, G. (1983). Endlichkeitssätze für abelsche Varietäten über Zahlkörpern. *Inventiones Mathematicae, 73*(3), 349–366.
- Faltings, G. (1984). Calculus on arithmetic surfaces. *Annals of Mathematics, 119*(2), 387–424.
- Frey, G. (1986). Links between stable elliptic curves and certain Diophantine equations. *Annales Universitatis Saraviensis, 1*, 1–40.
- Gross, B. H., & Zagier, D. B. (1986). Heegner points and derivatives of L -series. *Inventiones Mathematicae, 84*(2), 225–320.
- Hartshorne, R. (1977). *Algebraic geometry*. Springer.
- Hindry, M., & Silverman, J. H. (2000). *Diophantine geometry: An introduction*. Springer.
- Iwaniec, H., & Kowalski, E. (2004). *Analytic number theory*. American Mathematical Society.
- Lang, S. (1983). *Fundamentals of Diophantine geometry*. Springer-Verlag.
- Lang, S. (2002). *Algebra* (Rev. 3rd ed.). Springer.
- Mazur, B. (1977). Modular curves and the Eisenstein ideal. *Publications Mathématiques de l'IHÉS, 47*, 33–186.
- Mumford, D. (1977). Hirzebruch's proportionality theorem in the non-compact case. *Inventiones Mathematicae, 42*, 239–272.
- Ribet, K. A. (1990). On modular representations of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ arising from modular forms. *Inventiones Mathematicae, 100*(1), 431–476.
- Rudin, W. (1976). *Principles of mathematical analysis* (3rd ed.). McGraw–Hill.
- Saito, T. (1988). Conductor, discriminant, and self-intersection number of the relative canonical sheaf. *Duke Mathematical Journal, 57*(1), 151–173.
- Sarnak, P. (1990). *Some applications of modular forms*. Cambridge University Press.
- Serre, J.-P. (1987). Sur les représentations modulaires de degré 2 de $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. *Duke Mathematical Journal, 54*(1), 179–230.
- Silverman, J. H. (1986). *The arithmetic of elliptic curves*. Springer.
- Soulé, C. (1992). *Lectures on Arakelov geometry*. Cambridge University Press.

- Szpiro, L. (1981). Propriétés numériques du faisceau dualisant relatif. *Astérisque*, 86, 44–78.
- Zhang, S. (1993). Admissible pairing on a curve. *Inventiones Mathematicae*, 112(1), 171–193.